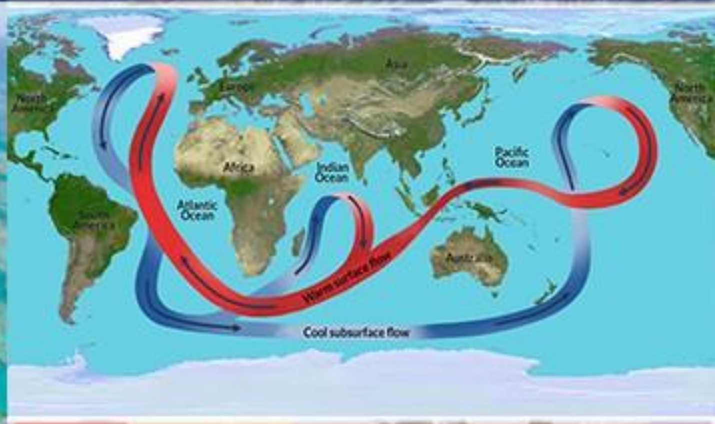
GEOGRAPHY

PHYSICAL GEOGRAPHY



OCEANOGRAPHY

MODULE - 7

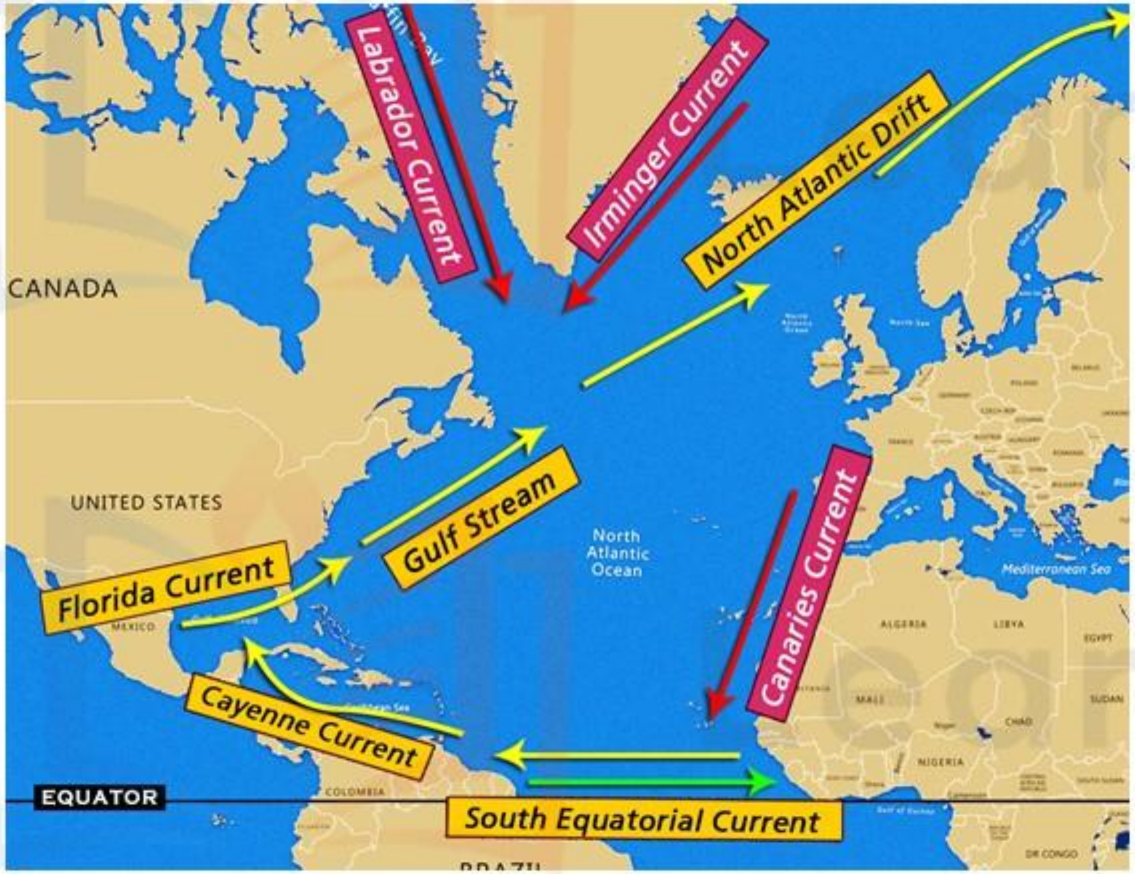
- CURRENTS
- CORAL REEFS

THE CIRCULATION OF THE ATLANTIC OCEAN

In the last module we discussed about the circulation in the Pacific Ocean. Let us now study more closely the circulation of ocean currents in the Atlantic Ocean.

1. We shall begin with the North and South Equatorial Current at the equator.
2. The steady Trade Winds constantly drift two streams of water from east to west.
3. At the "Shoulder" of North-East Brazil, the protruding land mass splits the ***South Equatorial Current*** into the ***Cayenne Current*** which flows along the Guiana coast, and the ***Brazilian Current*** which flows southwards along the east coast of Brazil.

THE CIRCULATION OF THE ATLANTIC OCEAN



4 In the North Atlantic Ocean, the ***Cayenne Current*** is joined and reinforced by the ***North Equatorial Current*** and heads north-westwards as a large mass of equatorial water into the Caribbean Sea.

5 Part of the current enters the Gulf of Mexico and emerges from the Florida Strait between Florida and Cuba as the ***Florida Current.***

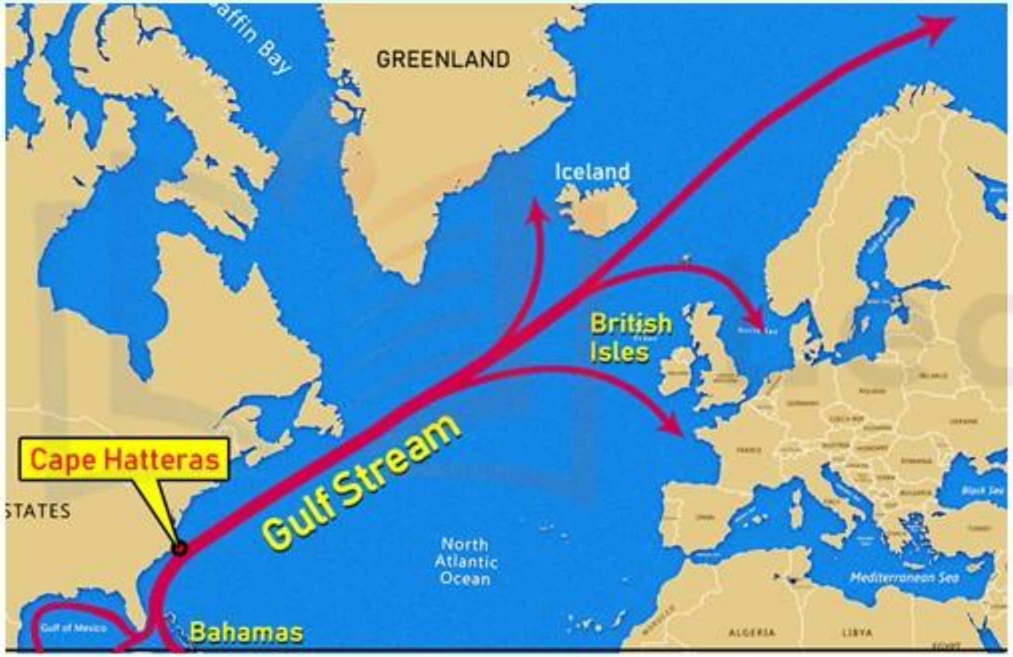
6 The rest of the equatorial water flows northwards east of the Antilles to join the ***Gulf Stream*** off the south-eastern USA.

THE CIRCULATION OF THE ATLANTIC OCEAN

SHORT NOTES ON GULF STREAM

7 The ***Gulf Stream*** drift is one of the strongest ocean currents- 35 to 100 miles wide, 2,000 feet deep and a velocity of 3 miles/hour.

8 The current hugs the coast of America as far as Cape Hatteras, where it is deflected eastwards under the combined influence of the Westerlies and the rotation of the Earth.

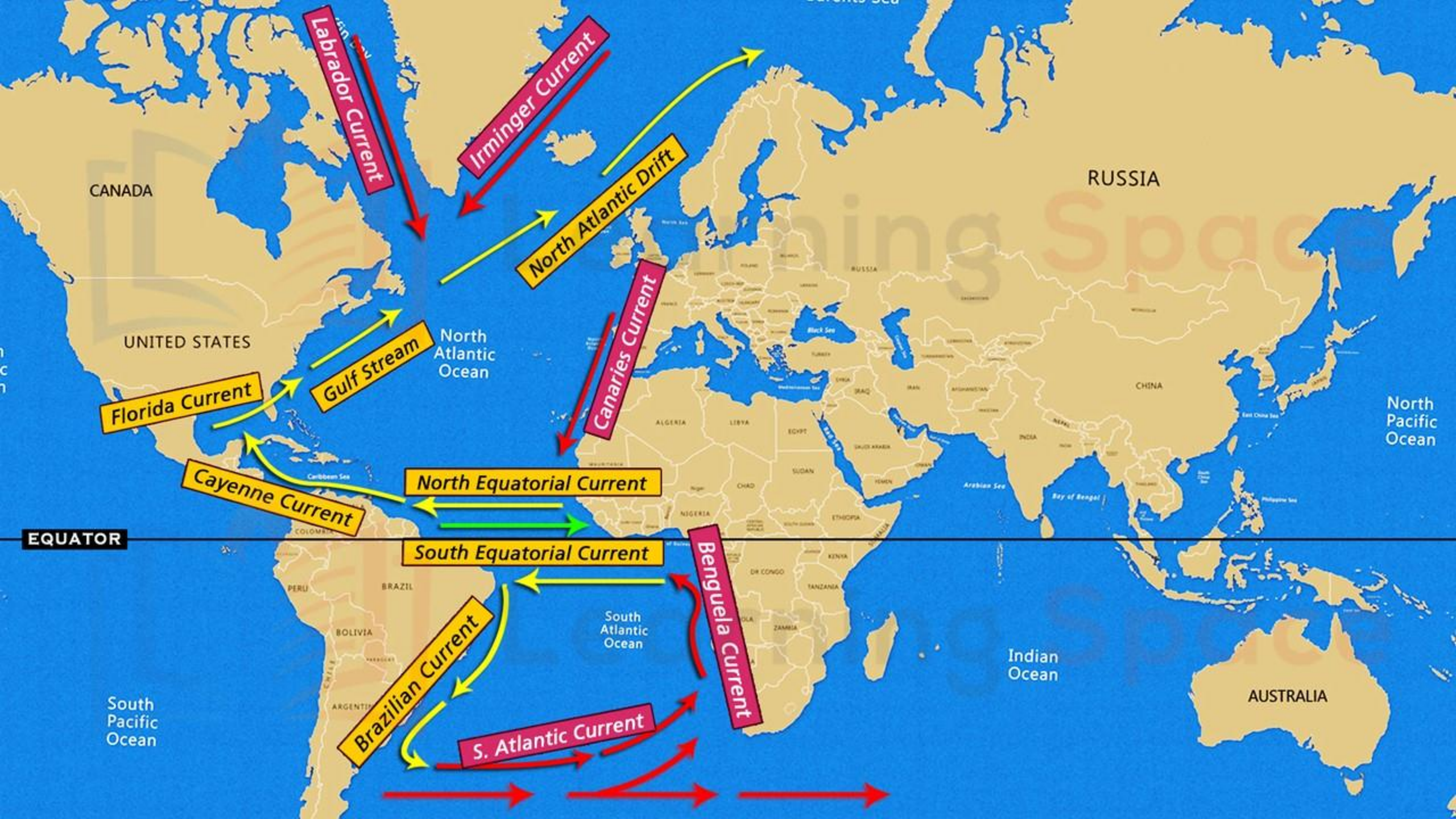


9 It reaches Europe as the ***North Atlantic Drift.*** This current, flowing at 10 miles per day, carries the warm equatorial water for over a thousand miles to the coasts of Europe.

10 From the North Atlantic, it fans out in three directions:

- eastwards to Britain,
- northwards to the Arctic and
- southwards along the Iberian coast, as the cool ***Canaries current***.

11 Oceanographic researchers show that almost two-thirds of the water brought by the Gulf stream to the Arctic regions is returned annually to the tropical latitudes by dense, cold polar water that creeps southwards in the ocean depths.



Labrador Current
Irminger Current
North Atlantic Drift
CANADA
RUSSIA
UNITED STATES
Gulf Stream
North Atlantic Ocean
Florida Current
Canaries Current
North Pacific Ocean
CHINA
Cayenne Current
North Equatorial Current
EQUATOR
South Equatorial Current
Benguela Current
Brazilian Current
South Atlantic Ocean
Indian Ocean
South Pacific Ocean
AUSTRALIA
S. Atlantic Current

THE CIRCULATION OF THE ATLANTIC OCEAN

SARGASSO SEA

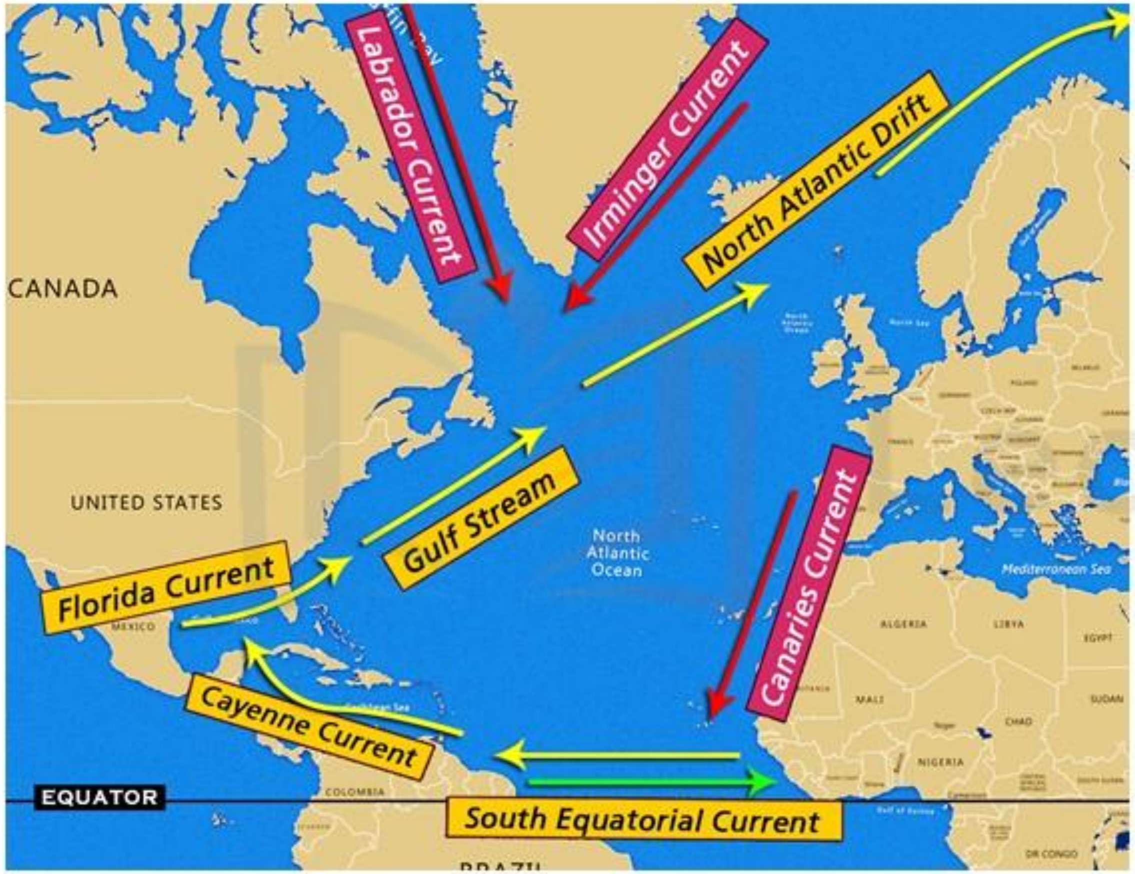
12 The Canaries Current flowing southwards eventually merges with the North equatorial Current, completing the clockwise circuit in the North Atlantic Ocean.

13 Within this ring of currents, an area in the middle of the Atlantic has no perceptible current.

14 A large amount of floating sea-weed gathers and the area is called the ***Sargasso Sea.***

15 This sea is covered with root-less sea weeds which obstruct navigation.

THE CIRCULATION OF THE ATLANTIC OCEAN



16 Apart from the clockwise circulation of the currents, there are also currents that enter the North Atlantic from the Arctic regions.

17 These cold waters are blown south by the out-flowing polar winds.

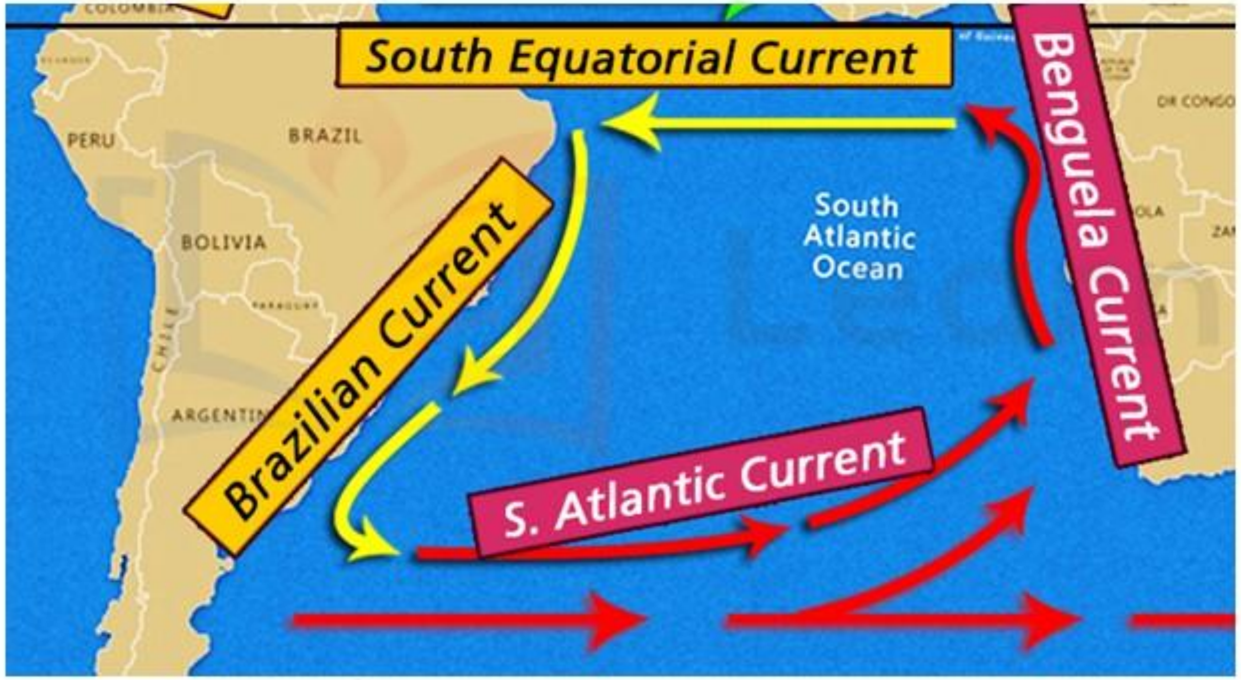
18 The ***Irminger current or East Greenland current*** flows between Iceland and Greenland and cools the North Atlantic Drift at the point of convergence.

THE CIRCULATION OF THE ATLANTIC OCEAN

Inter mixing of cold Labrador current and warm Gulf Stream near Newfoundland

19 The ***cold Labrador Current*** drifts south-eastwards between West Greenland and Baffin Island to meet the warm Gulf Stream Off Newfoundland, as far south as 50^0 N, where the icebergs carried south by the Labrador Current melt.

THE CIRCULATION OF THE ATLANTIC OCEAN



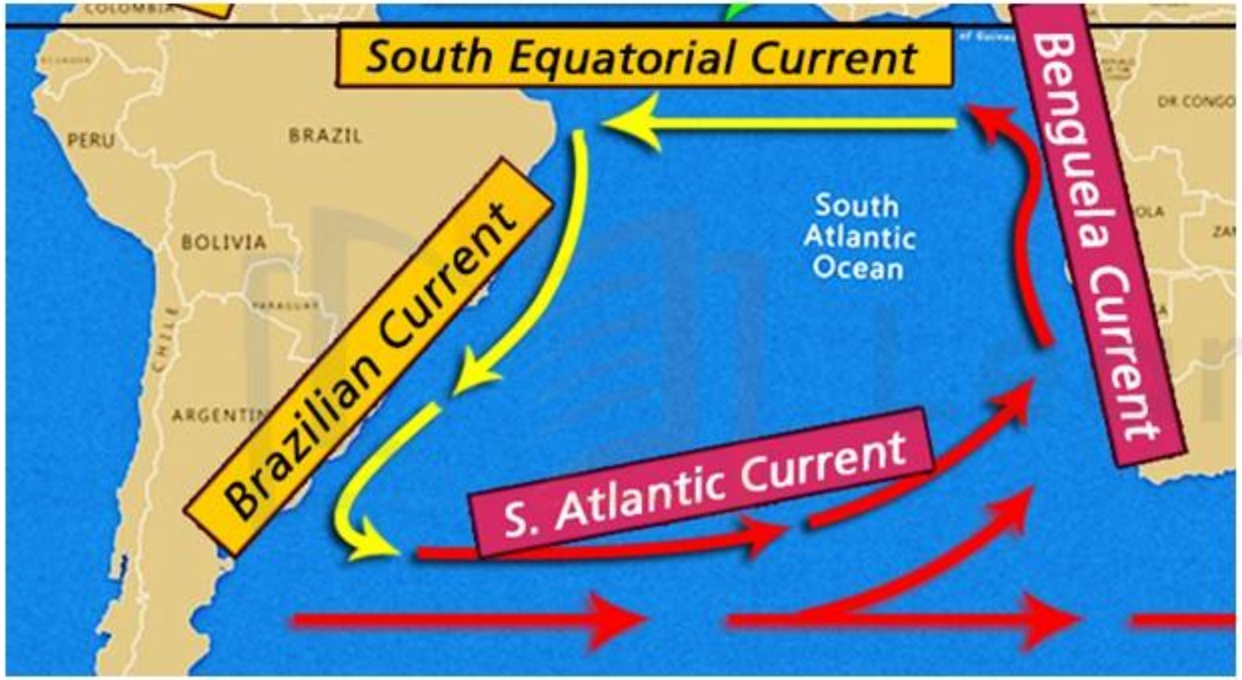
20 The South Atlantic Ocean follows the same pattern of circulation as the North Atlantic Ocean.

21 The major differences are that the ***circuit is anti-clockwise*** and the ***collection of sea-weed*** in the still waters of the mid-South Atlantic is not so distinctive.

22 Where the ***South Equatorial Current*** is split at Cape Sao Roque, one branch turns south as the warm ***Brazilian Current.***

23 Its deep blue waters are easily distinguishable from the yellow, muddy waters carried hundreds of miles out to sea by the Amazon further north.

THE CIRCULATION OF THE ATLANTIC OCEAN

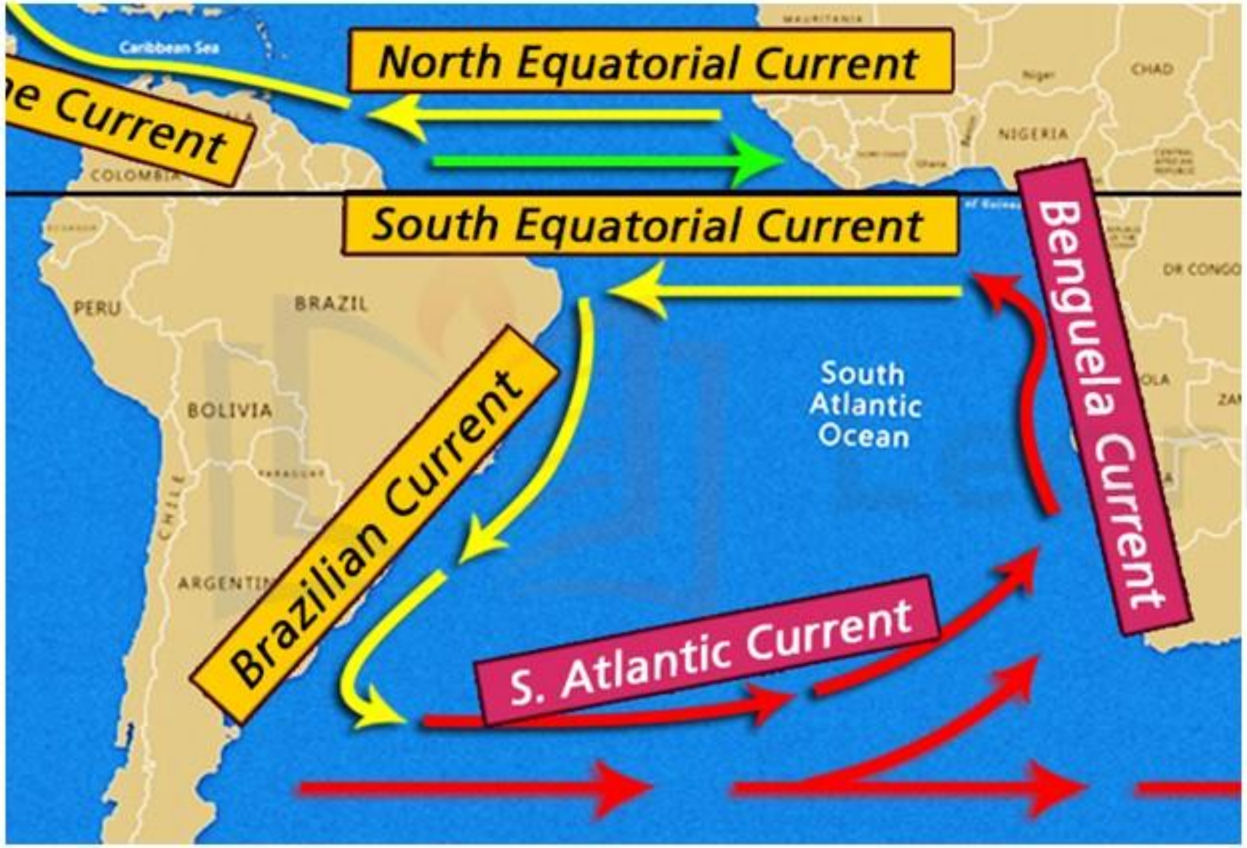


24 At about 40^0 S, the influence of the prevailing Westerlies and the rotation of the earth propel the current eastwards merging with the ***cold West Wind Drift*** as the ***South Atlantic Current.***

25 On reaching the west coast of Africa the current is diverted northwards as the cold ***Benguela Current*** (the counterpart of the Canaries Current).

26 It brings the cold polar waters of the ***West Wind Drift*** into the tropical latitudes.

THE CIRCULATION OF THE ATLANTIC OCEAN

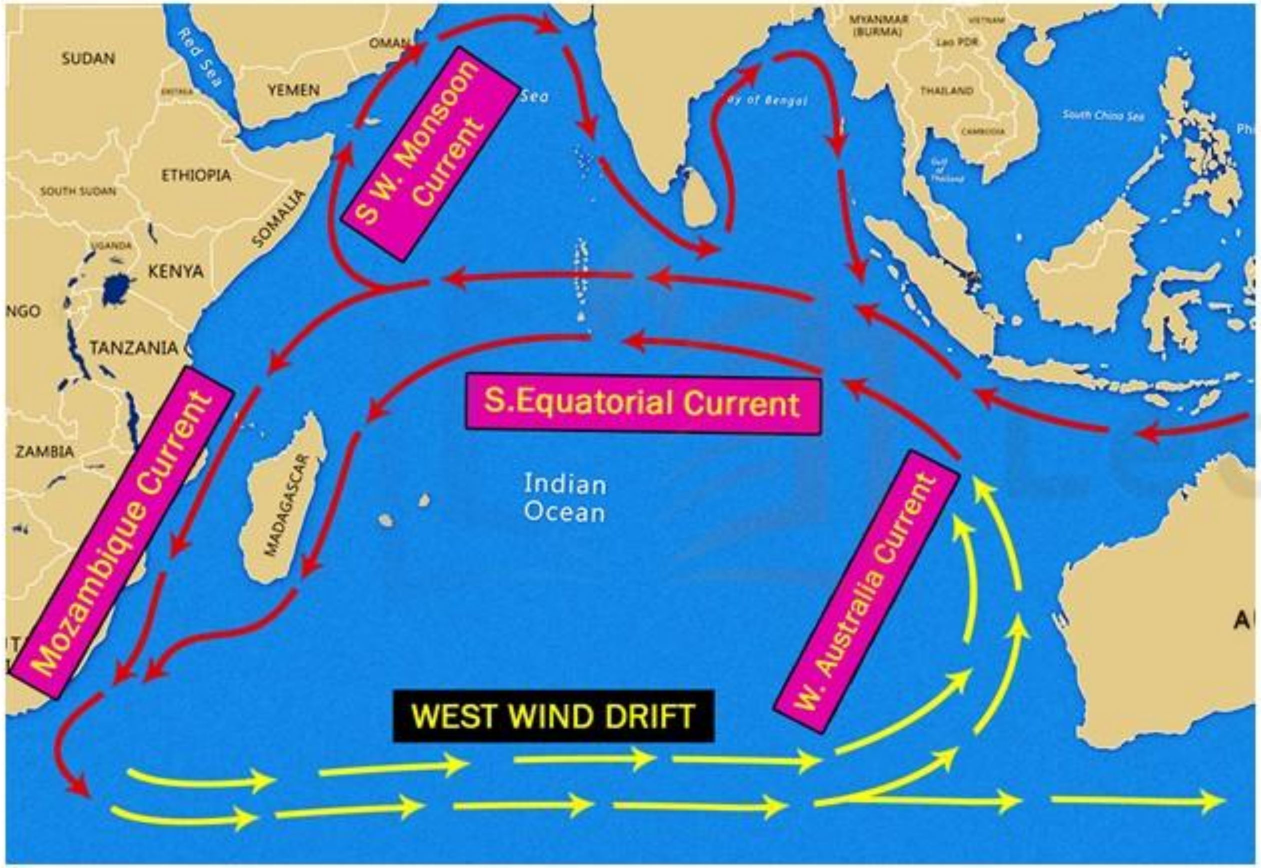


27 Driven by the regular South-East Trade Winds, the ***Benguela Current*** surges equatorwards in a north-westerly direction to join South Equatorial Current.

28 This completes the circulation of the currents in the South Atlantic.

29 Between the North and South Equatorial Currents is the east-flowing ***Equatorial Counter Current.***

THE CIRCULATION OF THE INDIAN OCEAN

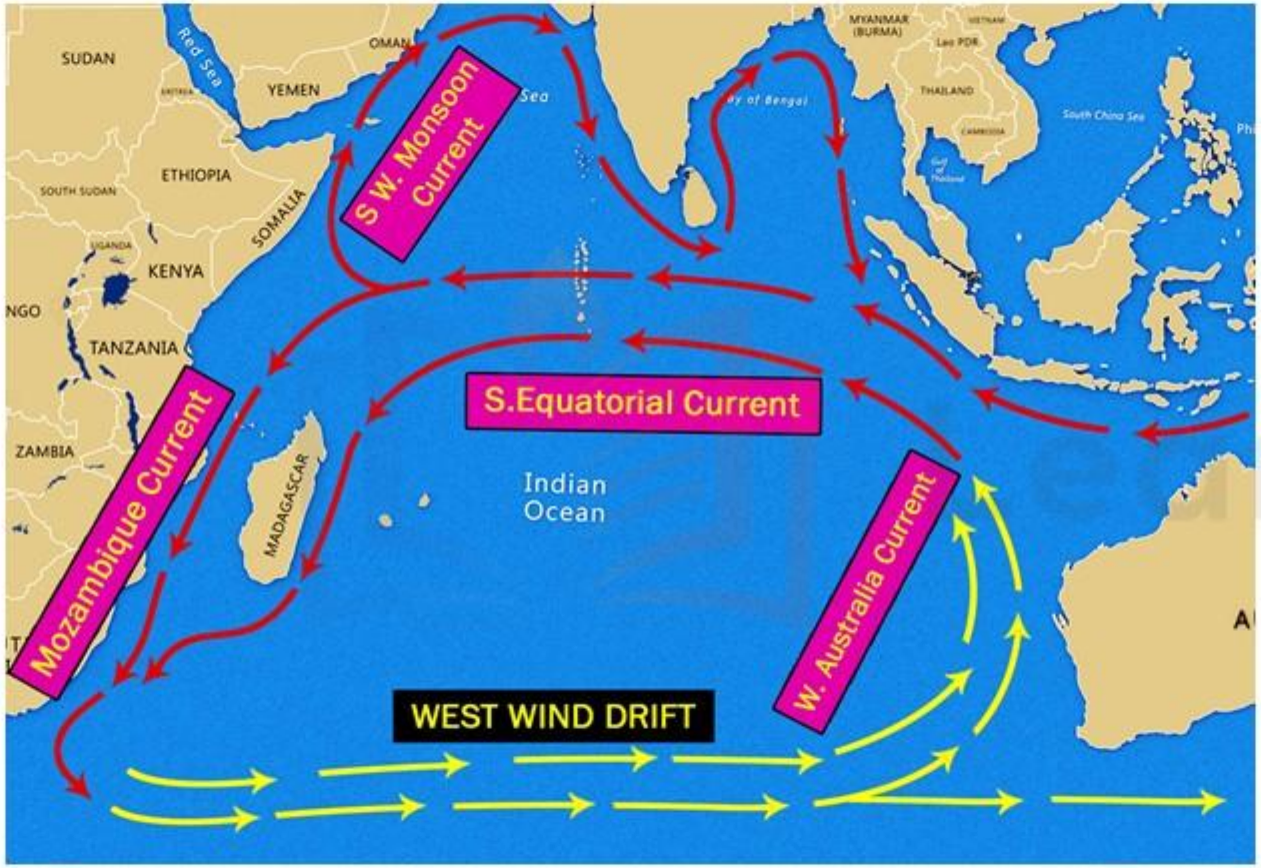


1 As in the other oceans, the currents of the South Indian Ocean form a circuit.

2 The Equatorial Current, turning southwards past Madagascar as the ***Agulhas or Mozambique Current*** merges with the West Wind Drift, flowing eastwards and turns equatorwards as the ***West Australian Current.***

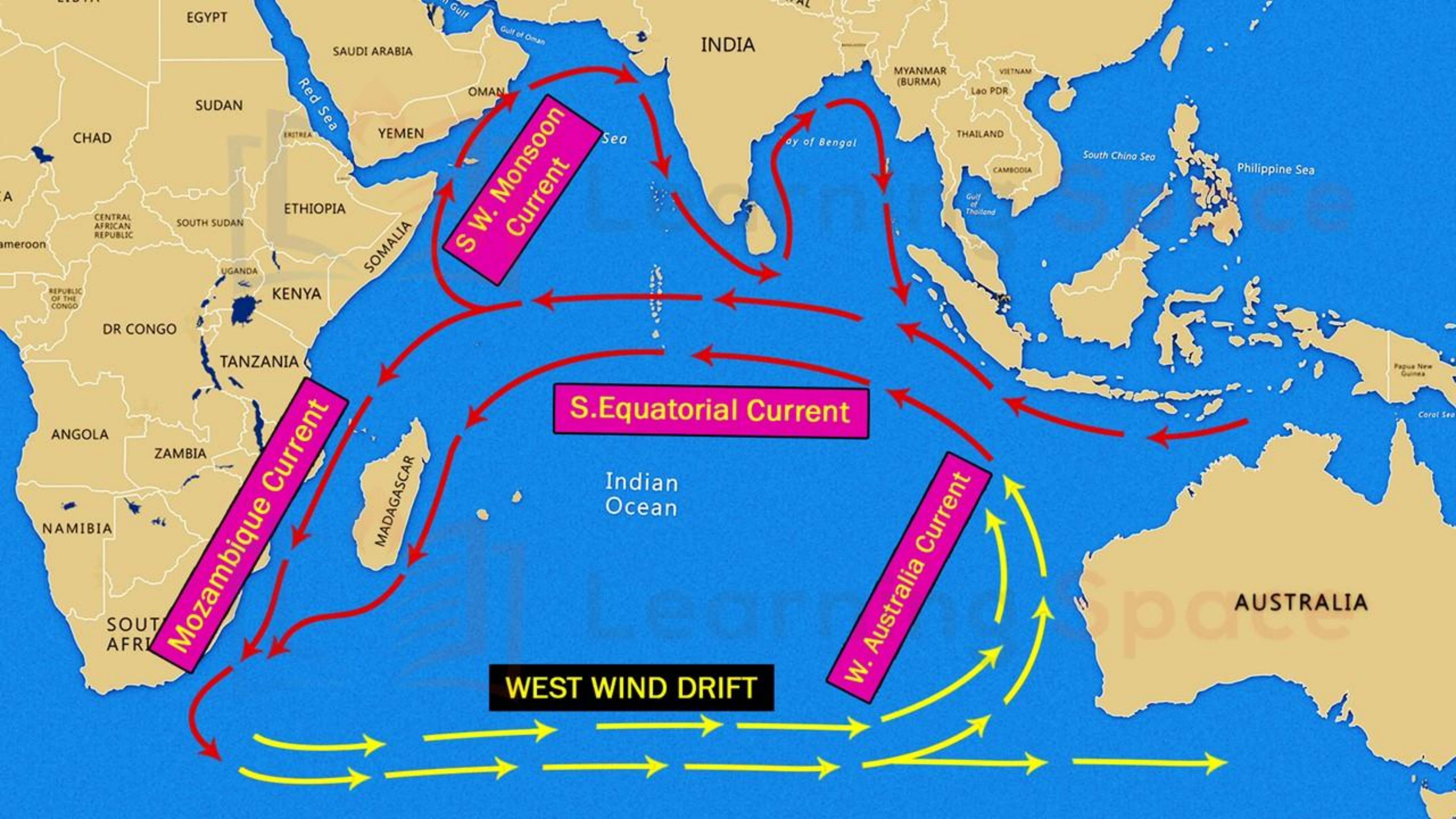
3 In the North Indian Ocean, there is a complete reversal of the direction of currents between summer and winter, due to changes of monsoon winds.

THE CIRCULATION OF THE INDIAN OCEAN



4 In summer from June to October, when there is the South-West Monsoon, the currents are blown from south-westerly direction as the ***South West Monsoon Drift.***

5 This is reversed in winter, beginning from December, when the North-East Monsoon blows the currents from the North-east as the ***North–East Monsoon Drift.***


S W. Monsoon Current
S.Equatorial Current
Mozambique Current
W. Australia Current
WEST WIND DRIFT
Indian Ocean
EGYPT
SAUDI ARABIA
INDIA
OMAN
SUDAN
CHAD
YEMEN
Red Sea
Gulf of Oman
ERITREA
ETHIOPIA
SOUTH SUDAN
CENTRAL AFRICAN REPUBLIC
SOMALIA
UGANDA
KENYA
REPUBLIC OF THE CONGO
DR CONGO
TANZANIA
ANGOLA
ZAMBIA
NAMIBIA
MADAGASCAR
SOUTH AFRICA
MYANMAR (BURMA)
Lao PDR
VIETNAM
THAILAND
CAMBODIA
Gulf of Thailand
Bay of Bengal
South China Sea
Philippine Sea
Papua New Guinea
Coral Sea
AUSTRALIA
Learning Space

OCEANOGRAPHY

CORAL REEF – AN INTRODUCTION

1 In tropical seas many kinds of coral animals and marine organisms such as coral polyps, calcareous algae, shell-forming creatures and lime-secreting plants live in large colonies.

2 Though they are very tiny creatures, their ability to secrete calcium carbonate within their tiny cells has given rise to a peculiar type of marine landform.

3 They exist in numerous species of many forms, colours and shapes.

4 Under favourable conditions, they grow in great profusion just below the water level.

5 Taking coral animals as a whole, the polyps are the most abundant and also the most important.

OCEANOGRAPHY

CORAL REEF – AN INTRODUCTION

When there is a disturbance to these conditions, coral bleaching will take place.

6. Each polyp resides in a tiny cup of coral and helps to form coral reefs.

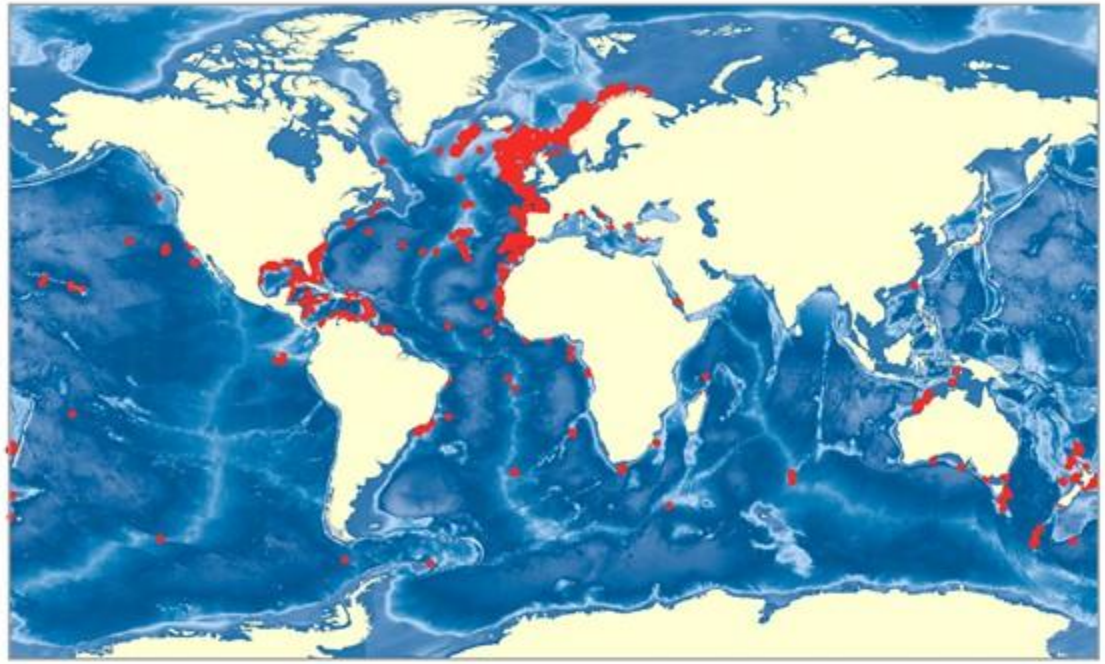
7. When they die, their limy skeletons are cemented into coralline limestone.

8. There are also non-reef-building species such as the 'Precious Corals' of the Pacific Ocean and the 'Red coral' of the Mediterranean which may survive in the colder and even the deeper waters.

9. As a general rule, they thrive well only in the warmer tropical seas.

OCEANOGRAPHY

CORAL REEF – AN INTRODUCTION



Reefs of the cold waters

DID YOU KNOW?

- Although fishermen and a few scientists have known about cold-water corals for nearly 250 years, it's only in the past few years that we've had the combination of advanced technology and political will to begin to explore them.
- Unlike tropical corals, cold-water corals don't have symbiotic algae living in their polyps so they don't need sunlight to survive. They feed solely by capturing food particles from the surrounding water. Their polyps tend to be much bigger than tropical corals.

WE EXPRESS OUR SINCERE THANKS FOR VIEWING THIS VIDEO

Presented by

Learning Space

For Suggestions:
suggestions@learningspace.in

To Contact us:
info@learningspace.in

OUR TEAM

G. V. Rao	Amrita Naidu
K. Srikanth	M. Binukrishna
D. Sunil Kumar	G. Ravi Babu
L. V. Krishna	D. Joji
Y. Deepthi	K. Victor Babu

Visit us at:
www.learningspacedigital.com
0866-24 44472
098499 42299